

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP5011
Subject Title	Machine Vision for Semiconductor Manufacturing and Inspections
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	To introduce machine vision for semiconductor manufacturing inspection such as microstructural inspection of wafers and IC packaging inspection.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand machine vision and its applications in the field of semiconductor industries; (b) utilize different lighting and optics techniques to acquire an image for digital processing and recognition; (c) study the machine vision solutions for Front-End, Middle-End, and Back-end wafer inspection. (d) understand the basic knowledge of methods of image processing, recognition, and understanding; (e) use general machine vision algorithms and software for image processing; (f) use deep learning models and software for defect inspection; (g) establish a system of machine vision for performing automatic inspection; and (h) apply machine vision in semiconductor manufacturing industries.
Subject Synopsis/ Indicative Syllabus	Overview of machine vision and its applications: Recent advancement of machine vision and its applications in semiconductor industries. Image acquisition: Components for image acquisition include lighting, optics, sensors, cameras, and scanners. Image digitization. The format and representations of digital images. Methodology of semiconductor Inspection: Front-end wafer inspection, backend assembly/package inspection, and mid-end wafer inspection for next-generation 3D IC. Image processing and analysis: Basic knowledge of methods of image processing, recognition, and understanding. Methods to extract information from digital images for further process. General machine vision algorithms. Basic knowledge of deep learning pipeline, including data preprocessing, learning, evaluation, and prediction. Machine vision implementation: Integration of the image acquisition components, machine vision algorithms/software, and control platform to perform automatic inspections for the required applications. Applications for semiconductor manufacturing and inspections: Configure for inspection tasks in the semiconductor manufacturing process such as sub-pixel alignment, identification, and flaw detection.

Teaching/Learning Methodology	Lecture: The fundamentals of machine vision and its applications for semiconductor manufacturing and inspection will be described. Students are free to request help. Students are encouraged to solve problems and to use their knowledge to verify their solutions before seeking assistance. Computer Laboratory: Students will write small programs for machine vision in groups to gain a deeper understanding of the topics discussed during the lectures. These sessions aim at developing the students’ intuition and critical thinking upon discussion and exchange of ideas to engage in the proposed activities.										
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)								
			a	b	c	d	e	f	g	h	
	1. Continuous Assessments	50%	✓	✓	✓	✓	✓	✓		✓	
	2. Project	50%	✓	✓	✓	✓	✓	✓	✓	✓	
	Total	100 %									
Continuous assessments, which include participation in the computer laboratory sessions, are used to assess the students’ understanding of basic concepts and principles in machine vision technologies for semiconductor processing applications. A group project is conducted to make a comprehensive assessment of students’ intended learning outcomes as stated above.											
Student Study Effort Expected	Class contact:										
	▪ Lecture							26 Hrs.			
	▪ Computer Laboratory							13 Hrs			
	Other student study efforts:										
	▪ Reading and self-study							81 Hrs.			
	The total student study effort							120 Hrs.			
Reading List and References	Zuech Nello, Understanding and Applying Machine Vision, Revised and Expanded, 2nd Edition, John Wiley & Sons, 2000 (https://fdocuments.in/reader/full/understanding-and-applying-machine-vision-569f4fdb6b4fd). E.R. Davies, Computer and Machine Vision: Theory, Algorithms, Practicalities, 4th Ed, Academic Press 2012. Milan Sonka, Vaclav Hlavac, Roger Boyle, Image processing, analysis and machine vision, 4th Ed, Cengage Learning, 2013 (https://kgut.ac.ir/useruploads/1550563201478ety.pdf). Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd Edition, Springer, 2021 (https://szeliski.org/Book/).										